WEEK 2 · DAY 1 · OVERVIEW

Changes in bp and Percent



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ABOUT 45 MIN WITH EXERCISES / 9 SECTIONS / 4 EXERCISES

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The Desk Lead's Note

Week 2 measures the dataset. The morning note opens with how far each series moved: on the day, over the week, over the month, and since the end of last year. Give me that for the 2, 5, 10 and 30 Yr and the two synthetic spreads as of 2 October, each in its right unit and tied to Excel. A market holiday must never show up as a zero move. Then show me what September's move in the 5 Yr did to a bond's price. Last, which days of 2026 moved the 10 Yr most? Describe; no views.

ABOUT 45 MIN WITH EXERCISES / 9 SECTIONS / 4 EXERCISES

Your Route Through Day 1

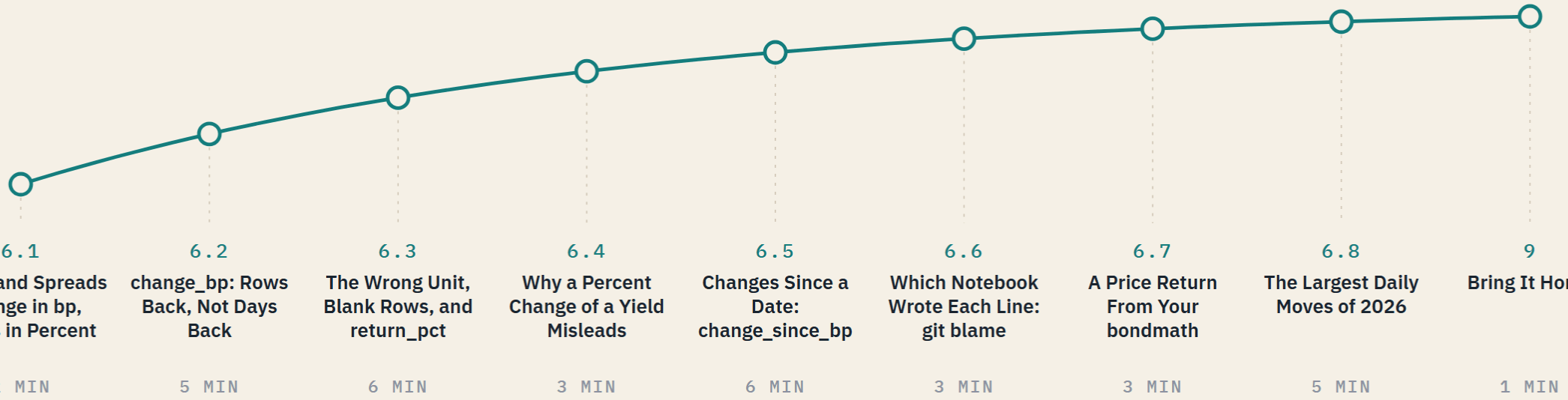


FIGURE 1 Nine sections, in order, with about how long each takes.

Time. About 34 minutes for the lesson, then about 12 for the four exercises.

6.1 Yields and Spreads Change in bp, Prices in Percent

Two units for one move: the change follows the unit of the series

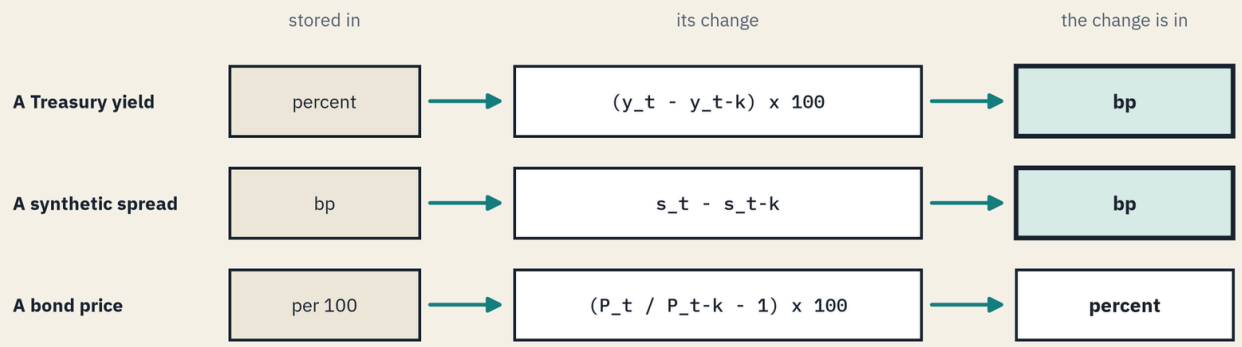


FIGURE 2 A yield moves in bp as a difference times 100, a spread in bp as a difference, a price in percent.

WHAT IT PRINTS

```
10 Yr: 5.24 to 5.28 percent, +4.0 bp
IG spread (synthetic): 98.0 to 97.0 bp, -1.0 bp
```

```
ten_move_bp = (ten.iloc[-1] - ten.iloc[-2]) * 100
ig_move_bp = ig.iloc[-1] - ig.iloc[-2]
print(f'10 Yr: {ten.iloc[-2]:.2f} to {ten.iloc[-1]:.2f} percent, {ten_move_bp:+.1f} bp')
print(f'IG spread (synthetic): {ig.iloc[-2]:.1f} to {ig.iloc[-1]:.1f} bp, {ig_move_bp:+.1f} bp')
```

A yield's move is a difference times 100, a spread's a plain difference, both in bp. Next, one function for both.

6.2 change_bp: Rows Back, Not Days Back

Rows back, not days back: change_bp(ten) in the week of Labor Day 2026

date	level, percent	shift(1)	change, bp	
Thu 3 Sep	4.77	4.79	-2	
Fri 4 Sep	4.78	4.77	+1	
Mon 7 Sep	Labor Day: no row, nothing to shift			
Tue 8 Sep	4.80	4.78	+2	from Friday
Wed 9 Sep	4.83	4.80	+3	

shift(1) moves each value down one row; the change is (level - shift(1)) x 100.

FIGURE 3 The 10 Yr from 3 to 9 September 2026: Labor Day has no row, so Tuesday's change is from Friday.

THE 10 YR'S MOVES

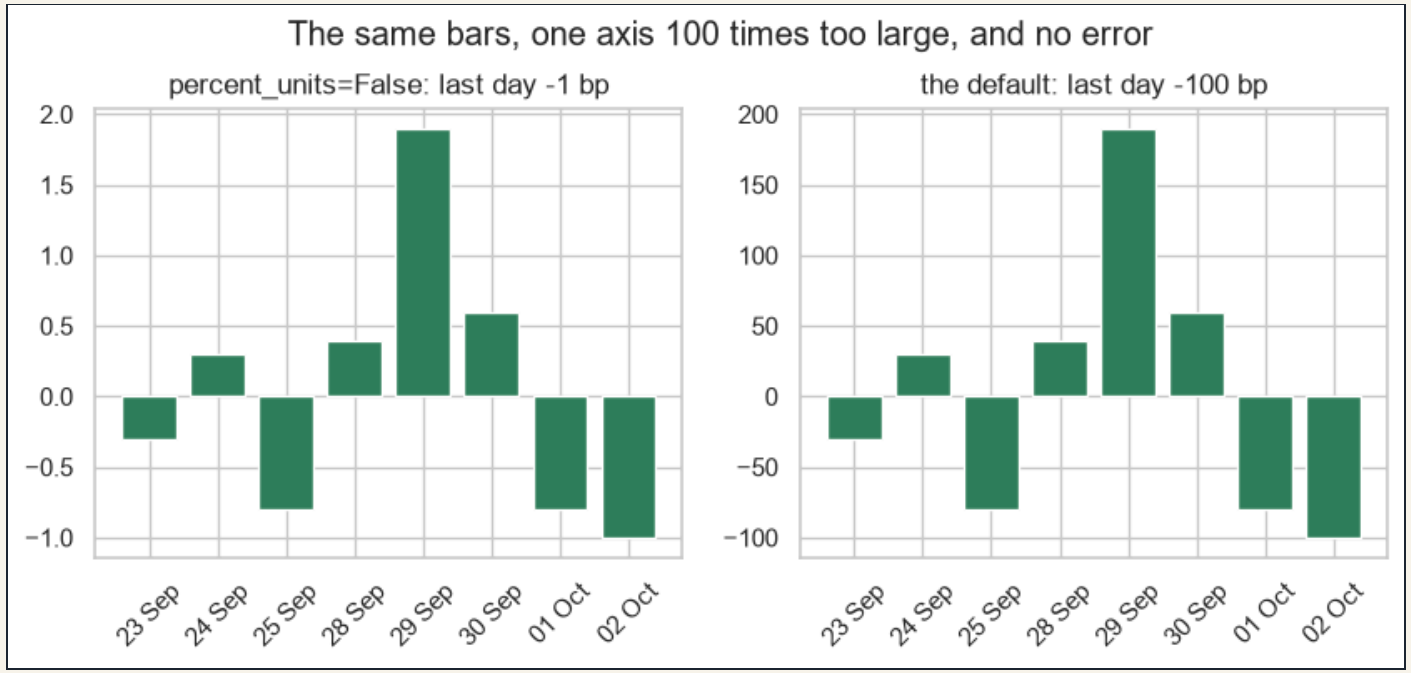
DATE	LEVEL_PCT	CHG_1_BP
2026-09-25	5.17	-1.0
2026-09-28	5.24	7.0
2026-09-29	5.26	2.0
2026-09-30	5.29	3.0

EXCEL SIDE BY SIDE

PYTHON	EXCEL
`md.change_bp(ten)`	`=(B3-B2)*100`

change_bp **counts rows, so a holiday is skipped, never a zero, and it equals Excel on every row of 2026. Next, the inputs it cannot fix by itself.**

6.3 The Wrong Unit, Blank Rows, and return_pct



THE CODE

```
ig_bp = md.change_bp(ig, percent_units=False)
ig_as_yield = md.change_bp(ig)

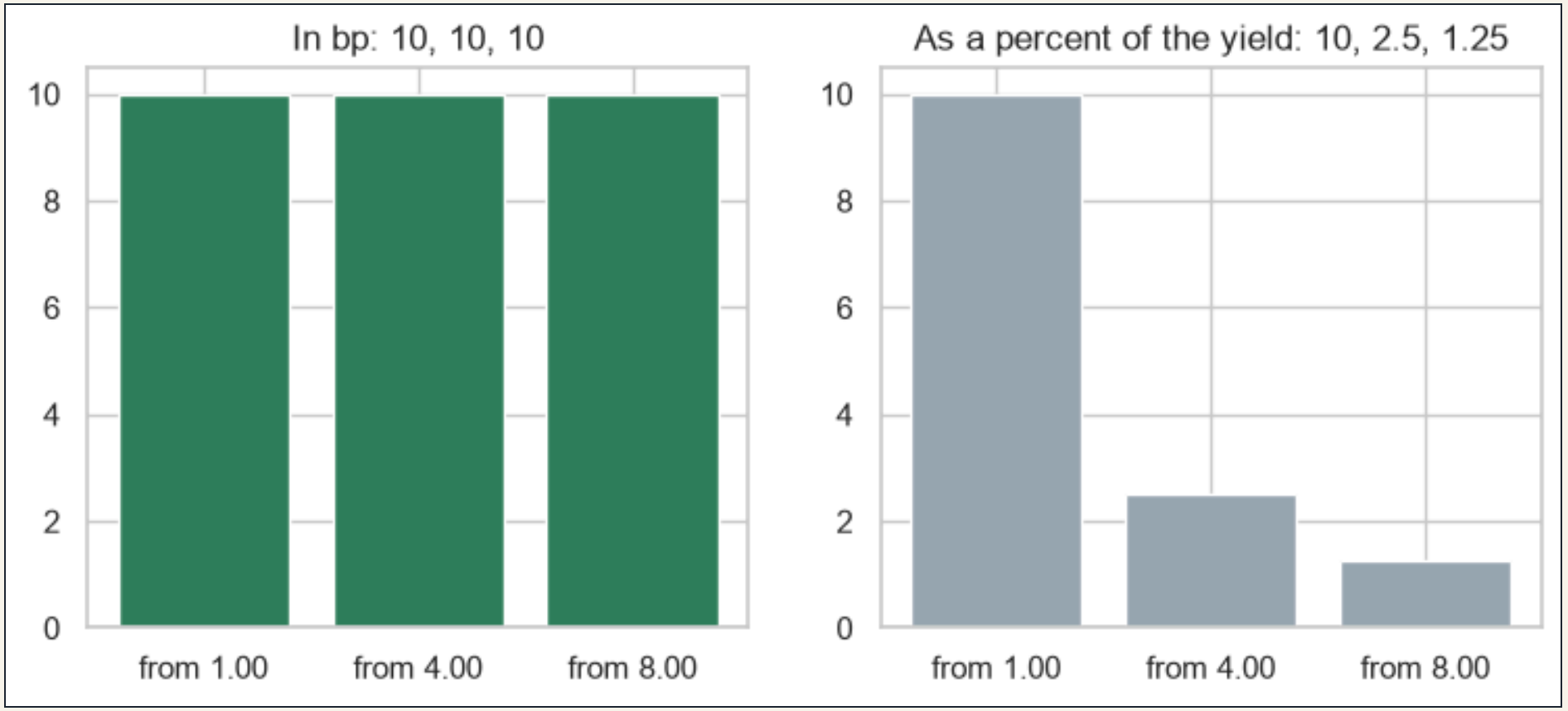
Last day: -1.0 bp, or -100.0 read as a yield
```

EXCEL SIDE BY SIDE

JOB	EXCEL
The spread's last change	`=D440-D439`
The same, read as a yield	`=(D440-D439)*100`

Pass `percent_units=False` for a spread, drop blank rows first, and use `return_pct` for a price. Now the trap the unit argument guards against.

6.4 Why a Percent Change of a Yield Misleads



The percent of a yield depends on where it started; the bp move does not. Yields and spreads are compared in bp. Next, a move since a date.

6.5 Changes Since a Date: change_since_bp

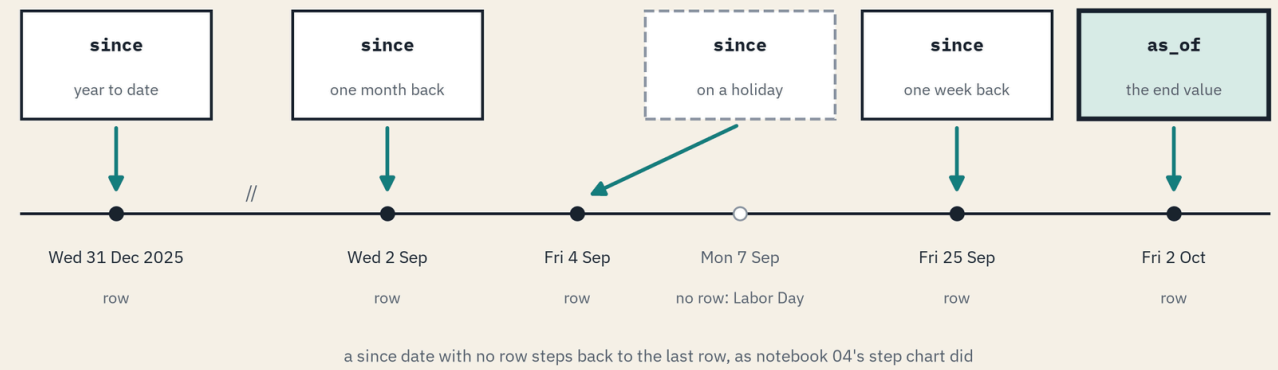


FIGURE 4 Since dates for 2 October 2026 land on rows, and Labor Day steps back to Friday 4 September.

WEEK, MONTH, YEAR TO DATE

```
week         from 2026-09-25 at 5.17: +11 bp
month        from 2026-09-02 at 4.79: +49 bp
year to date from 2025-12-31 at 4.18: +110 bp
```

EXCEL SIDE BY SIDE

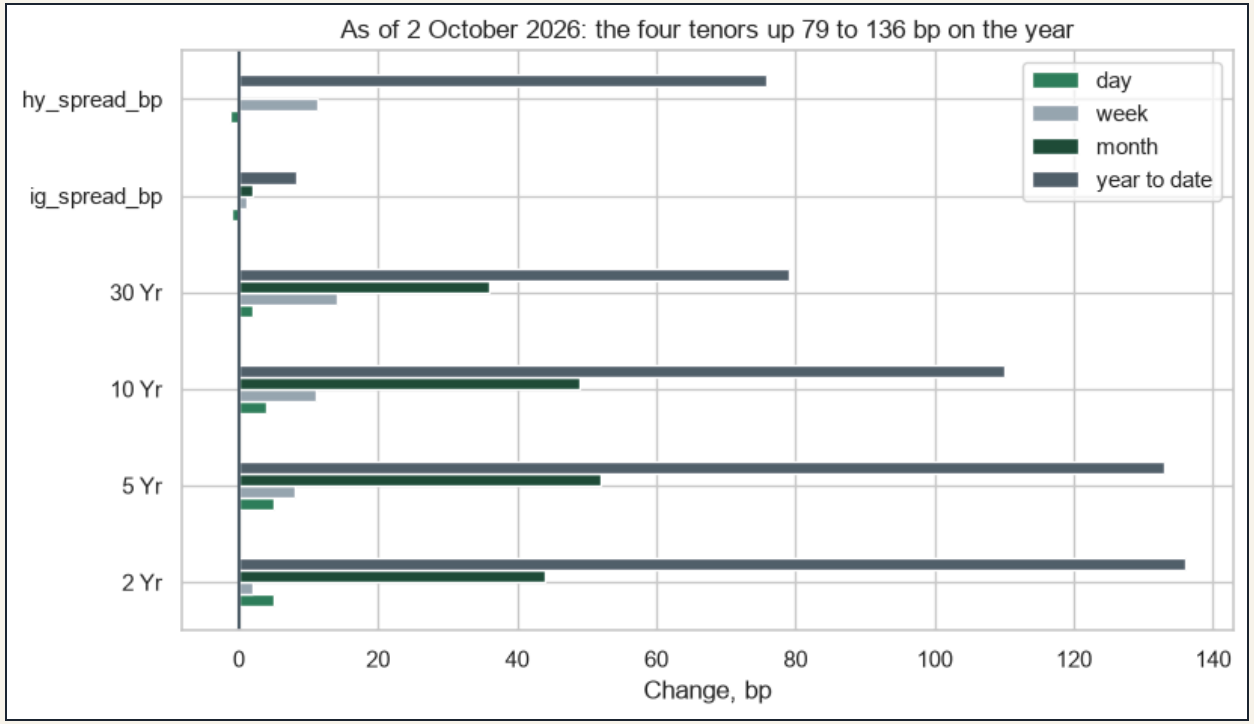
EXCEL
=(B251-XLOOKUP(EDATE(A251,-1),A:A,B:B,,-1))*100
=(B251-XLOOKUP(A251-7,A:A,B:B,,-1))*100

Two `value_as_of` calls and a subtraction.

The Morning Note's Table

EVERY MOVE, IN BP

	DAY	WEEK	MONTH	YEAR TO DATE
2 Yr	+5.0	+2.0	+44.0	+136.0
5 Yr	+5.0	+8.0	+52.0	+133.0
10 Yr	+4.0	+11.0	+49.0	+110.0
30 Yr	+2.0	+14.0	+36.0	+79.0
ig_spread_bp	-1.0	+1.1	+1.9	+8.2
hy_spread_bp	-1.3	+11.2	+0.1	+75.7



Every move in the morning note is in bp, from as-of values at both ends, equal to Excel on every 2026 row. The module is done for the day: commit it.

6.6 Which Notebook Wrote Each Line: git blame

COMMIT THE DAY'S WORK

```
!git -C "{work}" log --oneline --decorate
```

```
1f5838a (HEAD -> main) Notebook 06: change_bp, return_pct, and change_since_bp
ace289a (tag: week1) Notebook 05: write_cache, read_cache, and cached_fred_series
9f01b1c Notebook 04: last_per_period and value_as_of
```

WHICH COMMIT WROTE EACH LINE

```
lines = '-L "/def get_secret/,+1" -L "/def value_as_of/,+1" -L "/def change_bp/,+1"'
!git -C "{work}" blame -s --root {lines} marketdata.py
```

```
a28686d8  17) def get_secret(name: str, env_file: str | Path) -> str:
9f01b1c1 251) def value_as_of(series: pd.Series, as_of) -> tuple[pd.Timestamp, float]:
1f5838a2 339) def change_bp(series: pd.Series, periods: int = 1, percent_units: bool = True) -> pd.Series:
```

> `git blame` **names the commit behind each line, and so the reason for it. Now the bond the note asked about.**

6.7 A Price Return From Your bondmath

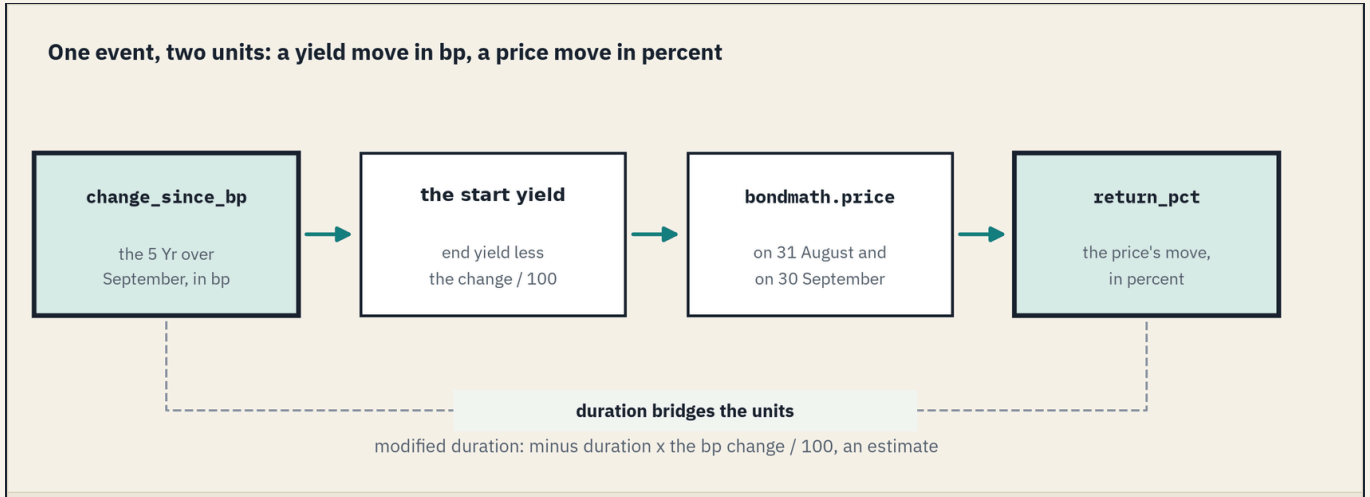


FIGURE 6 The 5 Yr's September move gives the bond's 31 August yield, two prices, and a return in percent.

SEPTEMBER'S MOVE IN THE 5 YR

```
5 Yr over September: +60 bp; the bond from 4.80 percent
```

PRICE IT ON BOTH DATES

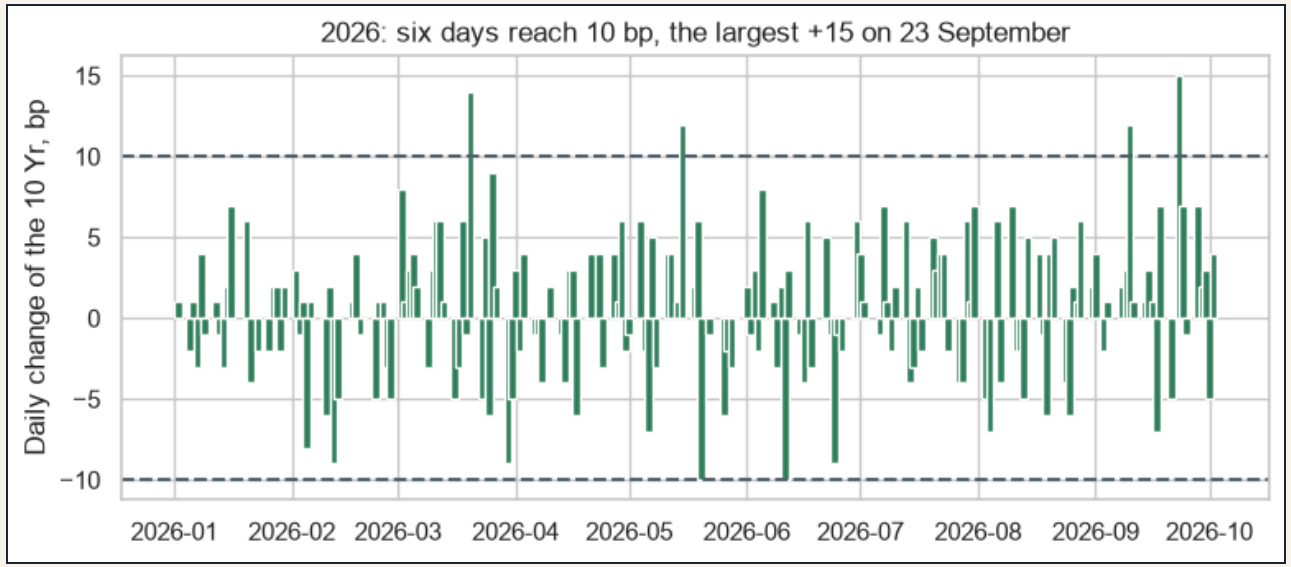
```
102.004270 to 99.350327: -2.6018 percent
```

EXCEL SIDE BY SIDE

JOB	EXCEL
The return in percent	`=(B2/B1-1)*100`

One event, two units: +60 bp on the yield, -2.60 percent on the price, and duration turns one into the other only because the move is in bp.

6.8 The Largest Daily Moves of 2026



THE FIVE LARGEST, WITH TIES

```
2026-09-23    15.0
2026-03-20    14.0
2026-05-15    12.0
2026-09-10    12.0
2026-05-20   -10.0
2026-06-11   -10.0
```

EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL	RESULT
The largest rise, and its date	`ten_2026.max()`, `.idxmax()`	`=MAX(C251:C440)`, `=XLOOKUP(MAX(C251:C440),C251:C440,A251:A440)`	15.000000000000036 on 2026-09-23

Six days: two tie at 10 bp, the fifth place.

Why Round First

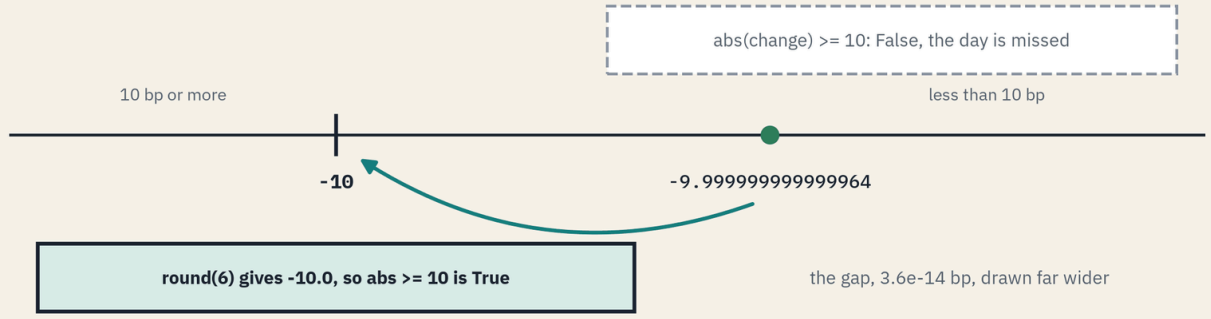


FIGURE 7 A change of -9.999999999999964 bp misses an abs of at least 10 until round(6) puts it on -10.

THE CODE

```
Unrounded: 4 days; rounded: 6 days
[-9.999999999999964, -9.999999999999964]
```

EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL	RESULT
Days of 10 bp or more	`(ten_2026.abs() >= 10).sum()`	`=SUMPRODUCT(--(ABS(C251:C440)>=10))`	4
The same, rounded	`(rounded.abs() >= 10).sum()`	`=SUMPRODUCT(--(ROUND(ABS(C251:C440),6)>=10))`	6

Round to 6 decimals before a count, a threshold or a rank.

Copy your work into your own folder.

Bring It Home

IN A POWERSHELL TERMINAL

```
# the work folder: use the name your setup cell printed
$work = "$env:TEMP\pfi_course3_06_k4w7q2xr"
Set-Location "$HOME\projects\my_bondmath"
Copy-Item "$work\marketdata.py", "$work\test_marketdata.py", "$work\course3_excel_reference.csv" .
python -m pytest -q test_marketdata.py
git diff --stat
git add marketdata.py test_marketdata.py
git commit -m "Notebook 06: change_bp, return_pct, and change_since_bp"
git blame -L "/def change_bp/,+1" marketdata.py
```

pytest shows 25 passed, and `git blame` names your new commit for the first line of `change_bp`.

You Can Now

- Say why yields and spreads change in bp and prices in percent.
- Measure a change k rows back with **`change_bp`**, for a yield in percent or a spread in bp.
- Explain why a percent change of a yield misleads.
- Measure a change since a date with **`change_since_bp`**, from as-of values at both ends.
- Price a bond on two dates with **`bondmath`** and put its return beside the yield change.
- Round before a count or a rank of computed changes.
- Find the commit behind a line with **`git blame -L`**.

NEXT
WEEK 2 · DAY 2

Rolling Windows and Volatility

How much a series has been moving: the sample standard deviation of daily changes in bp over a trailing window of rows, per day or labelled per year, tied to Excel and tested for lookahead.

